TT-I-549a

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SUPERSEDING
Fed. Spec. TT–I–549
December 9, 1930

FEDERAL STANDARD STOCK CATALOG

Section IV

(Part 5)

FEDERAL SPECIFICATION

FOR

INK; RED, WRITING

This specification was approved on the above date by the Director, Bureau of Federal Supply, for the use of all departments and establishments of the Government, and shall become effective not later than January 1, 1949, for use in procurement initiated on or after that date. It may be put into effect, however, at any earlier date after promulgation.

A. APPLICABLE SPECIFICATIONS

A–1. The following Federal Specifications, of the issues in effect on date of invitation for bids, shall form a part of this specification:

UU–P–121—Paper; Bond, White and Colored.

B. CLASSES

B–1. Red writing ink in accordance with this specification, shall be furnished in the following classes, as specified in the invitation for bids:

Class A.—Fluid.
Class B.—Powder.
Class C.—Tablets.

C. MATERIAL

C–1. The manufacturer is not restricted in a choice of dyes for making this ink. The quality of the ink shall be at least equal to that described in paragraph E–1.

D. GENERAL REQUIREMENTS

D–1. See section E.

E. DETAIL REQUIREMENTS

E–1. *Quality.*—The red ink as received (class A), or prepared by dissolving the powder (class B) or tablets (class C) in the amount of water recommended by the manufacturer, shall be a clear solution not inferior in any respect to one prepared by dissolving 5.5 grams of ciccein scarlet (British Colour Index No. 252), or 5.2 grams of azophloxine (British Colour Index No. 31), in one liter of distilled water. (See par. I–5.)

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E–2. *Properties.*—The sample shall be equal to the comparison ink (par. E–1) with respect to the following requirements: color of streaks when freshly made, permanence to light, and freedom from sediment before and after exposure to light and air. Writing formed with the ink shall photograph clearly on microcopying film. (See par. F–3.)

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F–1. *Sampling.*—

F–1a. *Fluid.*—An original unopened bottle bearing all of the manufacturers marks shall be submitted for test when the bottles contain not less than 4 fluid ounces. When the ink is furnished in smaller bottles, enough of them to aggregate at least 4 fluid ounces shall be submitted unopened and with all marks intact. The contents of these bottles shall be mixed to form a composite sample.

F–1b. *Powder or tablets.*—Enough of either the powder or tablets to make at least 16 fluid ounces of full strength ink shall be submitted for test in an original unopened container bearing all the manufacturer's marks.

F–2. *Preliminary examination.*—The powder or tablets shall be dissolved in the amount of distilled water recommended by the manufacturer. The ink of any class shall then be allowed to remain undisturbed in a stoppered glass container for 24 hours at temperatures in the range of 21° to 27°C. to allow any sediment to settle. The container shall then be held against the light and slowly tilted away from an observer who can thus see any sediment. Enough of the clear ink for all of the tests shall then be removed in a pipette.

F–3. *Tests.*—The sample shall be tested as follows in comparison with an ink made as described in paragraph E–1.

F–3a. *Ink streaks.*—Streaks shall be made by allowing measured portions of the ink, about 0.6 milliliter each, to flow freely across a sheet of white bond paper (Fed. Spec. UU–P–121, type III) which is pinned to a board or clamped to a pane of glass and held at an inclination of 45°. A streak of the ink undergoing test shall be alternated with a streak of the comparison ink on the same sheet of paper and four streaks of each ink shall be made in the machine direction of the paper. (See pars. I–3 and I–4.)

F–3b. *Form, color, and penetration.*—When the streaks are dry, the sheets shall be examined on the front and on the reverse sides. The streaks of the sample shall have the same general form and shall have approximately the same hue and intensity of color as those of the comparison ink. When viewed from the front and from the back, the streaks of the sample and those of the comparison ink shall be equally uniform in color and in degree of penetration.

F–3c. *Resistance to light.*—The paper shall be cut into strips 1 inch in width at right angles to the streaks. After the ink has been allowed to dry for at least 1 hour, the strips shall be used to make the following test: After exposure for 24 hours to the radiation from a glass-enclosed carbon arc such as the Fade-O-Meter type FDA–R or its equivalent, the streaks of the sample shall show no more evidence of fading than those of the comparison ink.

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F–3d. *Stability.*—Twenty-five milliliters of the sample and of the comparison ink shall be allowed to stand undisturbed in similar colorless glass crystallizing dishes of 2-inch inside diameter and about 1¾-inch depth, loosely covered to exclude dust. After 7-days exposure to diffused daylight and air at ordinary room temperature, the sample shall be as free from sediment and mold as the comparison ink.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G–1. *Packaging.*—Unless otherwise specified, commercial packages are acceptable under this specification. Ink in powdered or tablet form shall be packaged in siftproof containers. Tablets shall be protected from breakage in the container by two layers of cotton, one placed beneath and one above the tablets.

G–2. *Packing.*—Unless otherwise specified, the subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to the point of delivery called for in the contract or purchase order.

G–3. *Marking.*—Unless otherwise specified, shipping containers shall be marked with the name of the material, the size of the individual bottles, and the total number of bottles contained therein, as defined by the contract or order under which the shipment is made, the name of the contractor, and the number of the contract or order.

H. REQUIREMENTS APPLICABLE TO INDIVIDUAL DEPARTMENTS

H–1. The following specifications and other publications, of the issues in effect on date of invitation for bids, and special requirements, shall form a part of this specification, and shall be applicable to purchases made under this specification by the respective departments:

H–2. *Army.*—

H–2a. *Applicable specifications.*—

Federal Specification:

LLL–B–631—Boxes; Fiber, Corrugated (for Domestic-Shipment).

Joint Army-Navy Specifications:

JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.

JAN–P–125—Packaging and Packing for Overseas Shipment—Barrier-Materials, Waterproof, Flexible.

U. S. Army Specifications:

100–2 —Standard Specification for Marking Shipments by Contractors.

100–74—Packaging and Packing for Domestic Shipment of Glass Containers, One Gallon Capacity and Smaller.

Copies of specifications required for Army purchases may be obtained from the procuring agency. Both the title and identifying symbol or number should be stipulated when requesting copies.

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H–2b. *Packaging.*—

H–2b(1). *Class A, fluid.*—Four ounces, 1 pint, or 1 quart of writing ink shall be packaged in a narrow-mouth glass bottle closed with a knurled, curled-edge, continuous-thread metal screw cap fitted with a liner suitable for the product. The cap shall be coated outside and inside with a corrosion-resistant protective enamel and shall be tightened securely to prevent loosening and leakage while in transit.

H–2b(2). *Class B, powder.*—The correct amount of powder for reconstituting 1 quart of fluid ink, shall be packaged in an air-tight screw-cap glass bottle, the cap being further sealed by immersion in sealing wax. Five bottles shall then be packaged in a set-up box consisting of an outer sliding cover and inner holding tray or box made of not less than .024-inch-thick chipboard having inside dimensions approximately 3¾ inches long, 3⅜ inches wide, and ⅝ inch deep. Bottles shall be separated from each other by a snaked strip or scored sheet made from not less than .018-inch full bending chipboard.

H–2b(3). *Class C, tablets.*—Sixteen ink tablets, or a sufficient number of tablets to make 16 ounces of liquid ink, shall be packaged in a paperboard, wood, or metal cylindrical container with telescope cover, or any approved package which provides suitable structural strength and reclosure. The tablets shall not rattle excessively within the package.

H–2b(3)a. The package of tablets and sufficient cushioning material to minimize rattling, shall be enclosed in a 4-ounce plain round wide-mouth glass jar equipped with a curled-edge, continous-thread, metal screw cap having a water-resistant facing material suitable for the product. The metal cap shall be coated on both sides with a corrosion-resistant enamel.

H–2b(3)b. Twenty-four jars, each containing one package of ink tablets, shall be packed in a regular slotted box made of 200-pound-test double-faced corrugated fiberboard. The box shall have full-size top and bottom pads and a liner around the sides and ends made of 200-pound-test double-faced corrugated fiberboard. Full-size slotted partitions made from the same material shall be used to separate the jars. All flaps of each box shall be securely sealed by applying a good quality adhesive to all areas of contact between the flaps.

H–2c. *Packing.*—

H–2c(1). *For domestic shipment.*—

H–2c(1)a. *Class A, fluid.*—Four-ounce, 8-ounce, and 1-quart bottles shall be packed in accordance with the requirements of U. S. Army Specification 100–74. Only bottles of one size shall be packed in each shipping container.

H–2c(1)b. *Class B, powder.*—Two hundred bottles of powder, packaged as specified in paragraph H–2b(2), shall be packed in a fiberboard box conforming to Federal Specification LLL–B–631, style RSC. All flaps shall be securely sealed with a good quality adhesive applied throughout the entire area of contact between the flaps.

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H–2c(1)c. *Class C, tablets.*—Tablets, packaged as specified in paragraph H–2b(3), shall require no further packing for domestic shipment.

H–2c(2). *For overseas shipment.*—

H–2c(2)a. *Class A, fluid.*—Four-ounce, 1-pint, or 1-quart bottles of ink, packed as specified for domestic shipment (par. H–2c(1)a), shall be overpacked in a snug-fitting nailed wood box, style 2 or 4, constructed and strapped in conformance with the requirements of Joint Army-Navy Specification JAN–P–106. Only bottles of one size shall be packed in each shipping container.

H–2c(2)b. *Class B, powder.*—Two fiberboard boxes, each containing 200 bottles of powder (par. H–2c(1)b), shall be overpacked in a snug-fitting nailed wood box, style 2 or 4, constructed and strapped in conformance with the requirements of Joint Army-Navy Specification JAN–P–106.

H–2c(2)c. *Class C, tablets.*—Four boxes each containing 24 jars of tablets as specified in paragraph H–2b(3), shall be overpacked for overseas shipment in a snug-fitting nailed wood box, constructed and strapped in conformance with the requirements of Joint Army-Navy Specification JAN–P–106.

H–2c(2)d. *Liners.*—Each shipping box shall be provided with a sealed case liner made of waterproof barrier material conforming to the requirements of Joint Army-Navy Specification JAN–P–125, type L–2 or M. Gross weight of each shipping box shall not exceed 70 pounds.

H–2d. *Marking.*—

H–2d(1). *Packages.*—Each interior box of bottles shall be legibly labeled with the following information:

Stock number.
Nomenclature.
Quantity (net contents).
Directions for use (when applicable).
Name and address of manufacturer.

H–2d(2). *Shipping containers.*—The shipping containers shall be marked in conformance with the requirements of U. S. Army Specification 100–2.

H–3. *Navy.*—

H–3a. *Applicable specifications and other publications.*—

Federal Specifications:

NN–B–591—Boxes; Wood-Cleated-Fiberboard.
NN–B–601—Boxes; Wood-Cleated-Plywood, for Domestic Shipment.
NN–B–621—Boxes; Wood, Nailed and Lock-Corner.
NN–B–631—Boxes; Wood, Wire-Bound, for Domestic Shipment.
LLL–B–631—Boxes; Fiber, Corrugated (for Domestic Shipment).
LLL–B–636—Boxes; Fiber, Solid (for Domestic Shipment).

Joint Army-Navy Specifications:

JAN–P–105—Packaging and Packing for Overseas Shipment; Boxes, Wood, Cleated, Plywood.

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JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.
JAN–B–107—Boxes, Wood, Wirebound (Overseas Type).
JAN–P–108—Packaging and Packing for Overseas Shipment—Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.
JAN–P–125—Packaging and Packing for Overseas Shipment—Barrier-Materials, Waterproof, Flexible.
JAN–P–138—Packaging and Packing for Overseas Shipment—Boxes, Wood, Fiberboard-Lined.
JAN–P–140—Packaging and Packing for Overseas Shipment—Adhesives, Water-Resistant, Case-Liner.

Navy Department Specification:
General Specifications for Inspection of Material.

Bureau of Supplies and Accounts Publication:
Navy Shipment Marking Handbook.

Copies of Federal and Joint Army-Navy specifications (required for Navy purchases), Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained without cost upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va. When requesting specifications, refer to both title and number (or symbol).

H–3b. *Packaging.*—Unless otherwise specified, commercial packages are acceptable. Fluid ink shall be put up in commercial screw-capped bottles containing 1 pint (16 fluid ounces) or 4 fluid ounces, full measure, as specified in the contract or order. Bottles shall be packaged in substantial commercial containers provided with cardboard separators of sufficient thickness to withstand rehandling, to absorb jars, and to prevent contact between the bottles. Ink in powdered or tablet form shall be packed in siftproof containers.

H–3c. *Packing.*—

H–3c(1). *Domestic.*—Writing ink, packaged as described in paragraph H–3b, shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carrier, at the lowest rate, to the point of delivery. When wood-cleated fiberboard boxes, wood-cleated plywood boxes, nailed wood boxes, wire-bound wood boxes, corrugated fiber boxes, or solid fiber boxes are used they shall conform to the requirements of Federal Specifications NN–B–591, NN–B–601, NN–B–621, NN–B–631, LLL–B–631, and LLL–B–636, respectively. The gross weight of wood boxes shall not exceed 200 pounds; of fiberboard boxes, approximately 70 pounds.

H–3c(2). *Overseas.*—Writing ink, packaged as described in paragraph H–3b, shall be packed in cleated plywood boxes, nailed wood boxes, wire-bound wood boxes, fiberboard boxes, or fiberboard-lined wood boxes conforming to the requirements of Joint Army-Navy Specifications JAN–P–105, JAN–P–106, JAN–B–107, JAN–P–108, and JAN–P–138, respectively. Shipping containers shall be lined

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with a sealed waterproof bag, or its equivalent, made from material conforming to the requirements of Joint Army-Navy Specification JAN–P–125. The seams and closures shall be sealed with water-resistant adhesive conforming to the requirements of Joint Army-Navy Specification JAN–P–140. The gross weight of wood boxes shall not exceed 200 pounds; of fiberboard boxes, approximately 70 pounds.

H–3d. *Labeling and marking.*—

H–3d(1). *Labeling.*—All bottles shall bear a label showing the name of manufacturer or trade-mark.

H–3d(2). *Marking.*—In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with the Navy Shipment Marking Handbook.

H–3e. *Ordering information.*—Requests, requisitions, schedules, and contracts or orders should specify whether the subject commodity is to be packed and shipping containers marked for domestic or overseas shipment. (See pars. H–3c and H–3d.)

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I. NOTES

I–1. Purchasers should exercise any desired options offered herein and should specify the class of ink and size of unit required. (See pars. B–1, G–1, G–2, and G–3.)

I–2. The writing ink covered by this specification is intended for ordinary uses. It is not suitable for writing permanent records and is not resistant to water.

I–3. For making ink streaks (par. F–3a), a piece of glass tubing about 3.5 millimeters in bore and 250 millimeters long is used. A mark etched or scratched 62 millimeters from one end indicates the required volume, 0.6 milliliter, of ink. The ends of the tube can be fire polished, but should not be constricted. By drawing the ink up to the mark, holding the tube vertically against the paper near the top, and allowing the ink to flow suddenly down across the paper, uniform streaks will be obtained. Separate clean, dry tubes should be provided for the standard and the sample.

I–4. When streaks are made (par. F–3a), the ink flows smoothly down the sheet when it runs in the machine direction of the paper. If it runs across the machine direction, or grain, the paper warps into ridges which cause the ink to spread sideways and make uneven streaks. To determine the machine direction draw a line about 2 inches long on one corner of the sheet of paper. Cut a 1-inch square containing a portion of the line from the corner. Place the square on water and observe the curl. The parallel edges of the trough formed by the square as the paper curls away from the water indicate the machine direction.

I–5. It is common practice to refer to dyes by their numbers in one or both of the following books: Colour Index of the Society of Dyers and Colourists, or Gustav Schultz, Farbstofftabellen.

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I–6. It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

I–7. Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

I–8. An Index of Federal Specifications may be purchased as noted in the paragraph below, price to be obtained from the Superintendent of Documents.

I–9. Copies of this specification and of UU–P–121 may be obtained upon application, accompanied by postal note, money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Price 5 cents each.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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